

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

CO5 ASSESSMENT TOOL 2 – Seminar on Emerging Technologies

Course Code:	CSA1003	Course Title:	Software Engineering
CO Assessed:	CO5	Assessment:	Assessment Tool 2 – Seminar on Emerging Technologies
Weightage:	20%	Total Marks:	25
CO5 Statement:	Evaluate emerging technologies and societal impacts, maintaining and evolving software systems responsibly		
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Date:	24-08-2026	Duration:	60 minutes

Code of Conduct

I A.Akshaya(192425170) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Akshaya

Annexure A – Seminar on Emerging Technologies

Instructions to Students:

Students must select the topic **strictly according to the Serial Number (S.No.) order** assigned to them.

S.No.	Seminar Topic
1	Generative AI and Large Language Models (LLMs)
2	AI Agents and Autonomous AI Systems
3	Multimodal Artificial Intelligence
4	Explainable and Trustworthy AI
5	Edge AI and Intelligent Edge Computing
6	Federated Learning and Privacy-Preserving AI
7	Deep Learning for Computer Vision
8	AI-Powered Medical Image Analysis
9	Natural Language Processing with Transformer Models
10	Retrieval-Augmented Generation (RAG)
11	AI in Cybersecurity and Threat Detection
12	Zero Trust Architecture and Modern Cybersecurity
13	Post-Quantum Cryptography
14	Quantum Computing and Quantum Algorithms
15	Quantum Machine Learning

16	Cloud-Native Computing and Microservices
17	Serverless Computing and Function-as-a-Service
18	Edge Computing and Fog Computing
19	Cloud Security and Confidential Computing
20	Digital Twins and Intelligent Simulation
21	Internet of Things (IoT) and Smart Systems
22	Industrial Internet of Things (IIoT)
23	5G/6G Networks and Next-Generation Communication
24	Blockchain Technology and Decentralized Applications
25	Web3 and Decentralized Internet
26	Smart Contracts and Blockchain Applications
27	Augmented Reality (AR) and Virtual Reality (VR)
28	Extended Reality (XR) and Immersive Technologies
29	Robotics and Autonomous Robots
30	Human-Robot Interaction and Collaborative Robots
31	Autonomous Vehicles and Intelligent Transportation Systems
32	Smart Cities and AI-Driven Urban Technologies
33	Metaverse Technologies and Virtual Worlds
34	Brain-Computer Interfaces (BCI)
35	Wearable Computing and Smart Wearable Devices
36	Green Computing and Sustainable IT
37	AI for Software Engineering and Code Generation
38	DevSecOps and Intelligent CI/CD Pipelines
39	Digital Forensics and AI-Based Cybercrime Investigation
40	Privacy-Enhancing Technologies and Data Protection

Seminar Presentation Structure on **Robotics and Autonomous Robots**:

1. Introduction to the Emerging Technology

Robotics is an interdisciplinary field that combines mechanical engineering, electronics, computer science, artificial intelligence, and control systems to design machines that can perform physical tasks.

An **autonomous robot** is a robot that can sense its surroundings, make decisions, and perform actions with little or no continuous human control.

Unlike traditional robots that simply follow predefined commands, autonomous robots can **perceive, plan, adapt, and act** according to changes in their environment.

The technology is becoming increasingly important because robots can perform tasks in environments that may be dangerous, repetitive, inaccessible, or difficult for humans.

Examples include autonomous delivery robots, warehouse robots, agricultural robots, drones, self-driving vehicles, and planetary rovers.

2. Need and Motivation

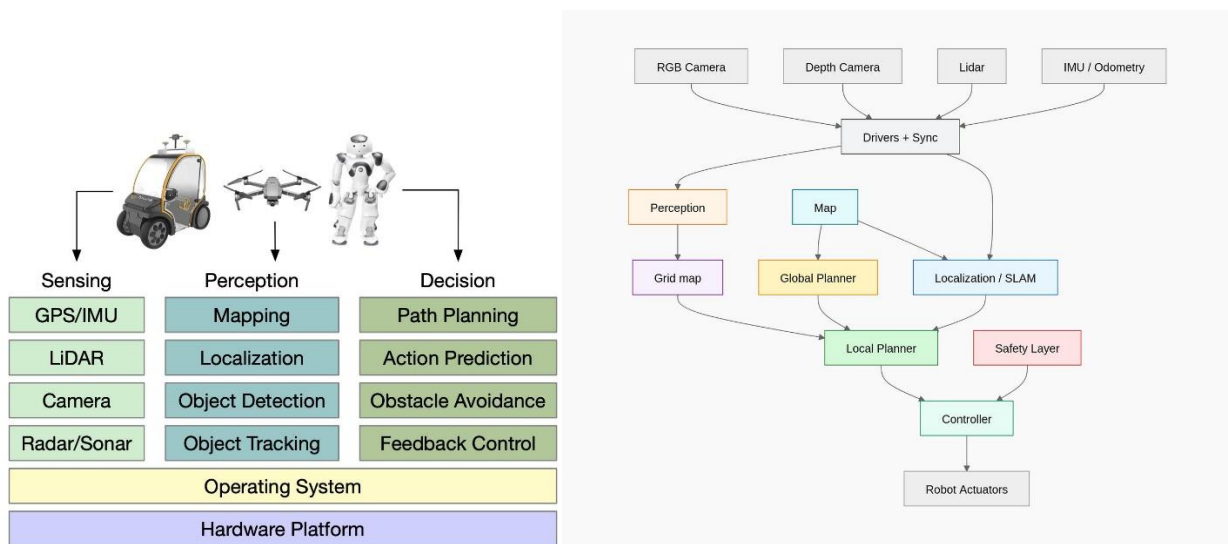
The main motivation for autonomous robotics is to reduce human effort while improving **efficiency, safety, accuracy, and productivity**.

Major needs:

- Perform dangerous tasks without exposing humans to risk.
- Automate repetitive activities.
- Operate in environments where humans cannot easily work.
- Reduce operational time and human effort.
- Improve accuracy in industrial and scientific tasks.
- Provide assistance in healthcare and daily activities.
- Explore environments such as oceans, mines, and other planets.
- Make real-time decisions based on environmental information.

For example, NASA's Perseverance rover uses autonomous navigation to detect hazards and select safer paths on Mars, reducing the need for Earth-based operators to specify every movement.

3. Working Principle / Architecture



1. Sensing: Cameras, LiDAR, ultrasonic sensors, GPS, and IMUs collect environmental information.

2. Perception: AI and computer vision process sensor data to identify objects, obstacles, people, roads, or other environmental features.

3. Localization: The robot estimates its current position.

4. Planning: A path-planning algorithm determines a suitable route to the destination.

5. Decision Making: The robot decides what action should be taken.

6. Control: Commands are sent to motors and actuators.

7. Feedback: Sensors continuously provide new information, allowing the robot to adjust its behavior.

4. **Key Components and Technologies**

The processing unit acts as the **brain of the robot**.

Examples:

- Raspberry Pi
- NVIDIA Jetson
- Arduino
- ESP32
- Industrial computers

C. Actuators

Actuators convert control signals into physical movement.

Examples:

- DC motors
- Servo motors
- Stepper motors
- Robotic arms

D. Artificial Intelligence

AI allows robots to make intelligent decisions.

Applications include:

- Object detection
- Object classification
- Path prediction
- Decision making
- Autonomous navigation

E. Computer Vision

Computer vision allows robots to understand images and videos.

Technologies include:

- OpenCV
- CNN
- YOLO
- Vision Transformers

F. SLAM

SLAM – Simultaneous Localization and Mapping allows a robot to build a map of an unknown environment while determining its own position within that map.

G. Path Planning

Algorithms such as **A***, Dijkstra's algorithm, RRT, and Dynamic Window Approach can be used to find suitable paths.

H. Robot Operating System

ROS/ROS 2 provides software libraries and tools for developing robotic applications, communication between components, sensor integration, and robot control.

5. Applications / Real-World Use Cases

Manufacturing

Robots can perform welding, assembly, inspection, packaging, and material handling.

Warehouses

Autonomous mobile robots can transport products and materials between different locations.

Agriculture

Robots can be used for:

- Crop monitoring
- Weed detection
- Fruit harvesting
- Spraying
- Soil analysis

Healthcare

Robots can assist with:

- Hospital delivery
- Patient assistance
- Surgical procedures
- Rehabilitation

Transportation

Autonomous technology is used in:

- Self-driving vehicles
- Autonomous shuttles
- Delivery robots
- Drones

Space Exploration

Autonomous robots are extremely useful for exploring planets where direct human control is difficult.

NASA's Perseverance can create terrain maps, detect hazards, and autonomously select safer routes.



6. Advantages and Limitations

Advantages

- Reduces human effort.
- Can operate continuously.
- Can work in dangerous environments.
- Improves productivity.
- Provides faster responses.
- Reduces repetitive human work.
- Can make real-time decisions.
- Supports exploration of inaccessible environments.

Limitations

- High development cost.
- Requires sophisticated hardware and software.
- Sensor errors can affect decisions.
- Battery life can be limited.
- Difficult to handle completely unknown environments.
- AI models may make incorrect predictions.
- Requires extensive testing for safety.
- Maintenance can be complex.

7. Recent Developments and Trends

1. AI-Powered Robots

Modern robots increasingly use AI and machine learning for perception, decision-making, and navigation.

2. Humanoid Robots

Humanoid robots are being developed to interact with environments designed for humans and perform multiple tasks.

3. Generative AI and Robotics

Generative AI is being explored for interpreting instructions, planning actions, and improving human-robot interaction.

4. Multi-Sensor Fusion

Robots increasingly combine information from cameras, LiDAR, IMUs, GPS, and other sensors to improve environmental understanding.

5. Autonomous Navigation

Robots are becoming better at navigating complex environments without continuous human instructions.

6. Autonomous Space Exploration

A particularly interesting recent development is NASA's **Mars Global Localization** system. In 2026, Perseverance demonstrated the ability to determine its location on Mars by matching panoramic camera imagery with orbital terrain maps. NASA reports that the system can locate the rover to roughly **25 cm** in about two minutes.

NASA has also reported the use of generative AI to help plan rover drive routes by selecting waypoints, further reducing workload on human operators.

8. Challenges and Future Scope

Challenges

Safety

Robots working around humans must make safe decisions and avoid collisions.

Perception

Changing lighting, weather, obstacles, and unfamiliar objects can make environmental understanding difficult.

Navigation

Robots may have difficulty navigating environments where maps or GPS are unavailable.

Power

Autonomous robots require sufficient battery capacity for sensing, computing, and movement.

Real-Time Processing

Robots often need to process sensor information and respond within milliseconds or seconds.

Reliability

A failure in the perception or control system can cause the robot to behave incorrectly.

Future Scope

Future autonomous robots are expected to become:

- More intelligent.
- More energy efficient.
- Better at human interaction.
- More capable of learning from experience.
- Better at navigating unknown environments.
- More capable of cooperating with other robots.
- More useful in healthcare, agriculture, logistics, disaster management, and space exploration.

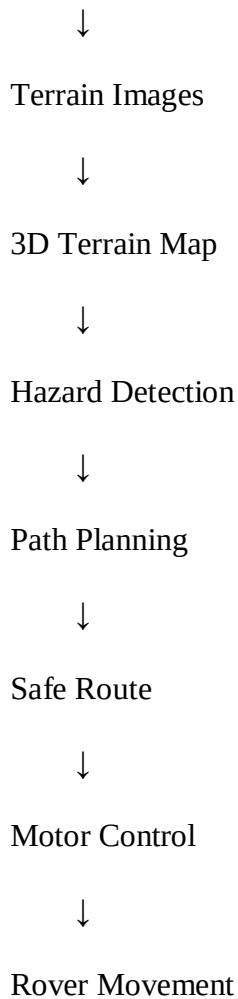
9. Case Study / Example

Case Study: Autonomous Navigation of NASA's Perseverance

NASA's **Perseverance rover** is an excellent real-world example of autonomous robotics.

The rover operates on Mars, where communication with Earth has significant delays. Therefore, it cannot depend on humans to control every movement.

Navigation Cameras



10. Conclusion and References

Robotics and Autonomous Robots are important emerging technologies that combine mechanical systems, electronics, artificial intelligence, computer vision, sensors, and control systems.

Traditional robots mainly follow predefined instructions, whereas autonomous robots can **sense their surroundings, make decisions, plan actions, and adapt to changes**.

Their applications range from manufacturing and agriculture to healthcare, transportation and space exploration. However, challenges such as safety, reliability, power consumption, environmental uncertainty, and real-time processing still need to be addressed.

The development of systems such as NASA Perseverance demonstrates the potential of autonomous robotics. Its autonomous navigation and newer self-localization capabilities show how robots can increasingly perform complex tasks with less human intervention.

Therefore, autonomous robotics has strong future potential and is likely to become an important technology in **industry, transportation, healthcare, exploration, and everyday life**.

Rubrics for Calculation

Evaluation Criteria	Excellent (4 Marks)	Good (3 Marks)	Satisfactory (2 Marks)	Needs Improvement (1 Mark)
Technical Content & Subject Knowledge (25%)	Comprehensive and accurate content; demonstrates excellent understanding of the emerging technology	Good technical understanding with minor gaps	Basic understanding with limited technical depth	Incomplete content or major technical errors
Organization & Presentation Structure (15%)	Excellent logical flow with clear introduction, content, conclusion, and effective transitions	Well-structured with minor organizational issues	Adequately structured but some sections lack clarity	Poorly structured and difficult to follow
Examples, Diagrams & Case Study (20%)	Excellent use of relevant diagrams, examples, applications, and real-world case studies	Good use of relevant examples and diagrams	Limited examples/diagrams with basic explanation	Inappropriate or insufficient examples/diagrams
Communication & Presentation Skills (20%)	Very clear, confident, engaging delivery with excellent eye contact, voice modulation, and time management	Clear and confident presentation with minor issues	Understandable but lacks confidence or engagement	Unclear delivery, excessive reading, or poor time management
Question Handling & Overall Performance (20%)	Answers questions accurately and confidently; demonstrates strong overall knowledge	Answers most questions correctly	Answers basic questions with some difficulty	Unable to answer questions or demonstrate adequate understanding

Criteria	Mark
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Technical Content & Subject Knowledge (25%)	/5
Organization & Presentation Structure (15%)	/5
Examples, Diagrams & Case Study (20%)	/5
Communication & Presentation Skills (20%)	/5
Question Handling & Overall Performance (20%)	/5
TOTAL	/25